

Problem

Given an amount and the denominations of coins available, determine how many ways change can be made for amount. There is a limitless supply of each coin type.

Example

There are ways to make change for : ,
 , and .

Submissions

Function Description

Complete the getWays function in the editor below.

getWays has the following parameter(s):

- int n: the amount to make change for
- int c[m]: the available coin denominations

Returns

- int: the number of ways to make change

Leaderboard

Input Format

The first line contains two space-separated integers and , where:

 is the amount to change

 is the number of coin types

Discussions

The second line contains space-separated integers that describe the values of each coin type.

Constraints

-
-
-
- Each is guaranteed to be distinct.

Hints

Solve overlapping subproblems using [Dynamic Programming \(DP\)](#):

You can solve this problem recursively but will not pass all the test cases without optimizing to eliminate the [overlapping subproblems](#). Think of a way to store and reference previously computed solutions to avoid solving the same subproblem multiple times. * Consider the degenerate cases:

- How many ways can you make change for cents? -

How many ways can you make change for cents if you have no coins? * If you're having trouble defining your solutions store, then think about it in terms of the base case . - The answer may be larger than a -bit integer.

Sample Input 0

Change Theme

Language

Python 3



```
1 import math
2 import os
3 import random
4 import re
5 import sys
6 def getWays(n, c):
7     dp = [0] * (n + 1)
8     dp[0] = 1
9     for coin in c:
10         for amount in range(coin, n + 1):
11             dp[amount] += dp[amount - coin]
12     return dp[n]
13 if __name__ == '__main__':
14     fptr = open(os.environ['OUTPUT_PATH'], 'w')
15
16     first_multiple_input = input().rstrip().split()
17
18     n = int(first_multiple_input[0])
19     m = int(first_multiple_input[1])
20
21     c = list(map(int, input().rstrip().split()))
22
23     ways = getWays(n, c)
24
25     fptr.write(str(ways) + '\n')
26     fptr.close()
27
```

Line: 27 Col: 1



Upload Code as File

Run Code

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Test against custom input